

81. Let  $y = y(x)$  be the solution curve of the differential equation  $\frac{dy}{dx} = \frac{y}{x}(1 + xy^2(1 + \log_e x))$ ,  $x > 0$ ,

$y(1) = 3$ . Then  $\frac{y^2(x)}{9}$  is equal to :

माना अवकल समीकरण  $\frac{dy}{dx} = \frac{y}{x}(1 + xy^2(1 + \log_e x))$ ,  $x > 0$ ,  $y(1) = 3$  का हल  $y = y(x)$  है। तो  $\frac{y^2(x)}{9}$  बराबर है :

(A)  $\frac{x^2}{7 - 3x^2(2 + \log_e x^2)}$

(B)  $\frac{x^2}{2x^3(2 + \log_e x^3) - 3}$

(C)  $\frac{x^2}{3x^3(1 + \log_e x^2) - 2}$

(D)  $\frac{x^2}{5 - 2x^3(2 + \log_e x^3)}$

**Ans.** D

**Sol.**  $\frac{dy}{dx} = \frac{y}{x}(1 + xy^2(1 + \log_e x))$ ,  $x > 0$

$$\frac{dy}{dx} = \frac{y}{x} + y^3(1 + \log_e x)$$

$$\frac{1}{y^3} \frac{dy}{dx} = \frac{1}{x y^2} + (1 + \log_e x)$$

$$\frac{1}{y^3} \frac{dy}{dx} - \frac{1}{x} \frac{1}{y^2} = (1 + \ln x)$$

Let  $\frac{1}{y^2} = t$

$$-\frac{2}{y^3} \frac{dy}{dx} = \frac{dt}{dx}$$

$$\therefore -\frac{1}{2} \frac{dt}{dx} - \frac{t}{x} = (1 + \ln x)$$

$$\frac{dt}{dx} + \frac{2t}{x} = -2(1 + \ln x)$$

if  $e^{\int \frac{2}{x} dx} = x^2$

$$\therefore tx^2 = \int -2(1 + \ln x) x^2 \cdot dx$$

$$tx^2 = -2 \left[ (1 + \ln x) \frac{x^3}{3} - \int \frac{x^2}{3} dx \right] + C$$

$$\frac{x^2}{y^2} = -2 \left[ \frac{x^3}{3} (1 + \ln x) - \frac{x^3}{9} \right] + C \quad \dots(1)$$

$$\because y(1) = 3$$

$$\frac{1}{9} = -2 \left[ \frac{1}{3} (1 + 0) - \frac{1}{9} \right] + C$$

$$C = \frac{5}{9}$$

$$\text{put } C = \frac{5}{9}$$

$$\Rightarrow \frac{y^2(x)}{9} = \frac{x^2}{5 - 2x^3(2 + \log_e x^3)}$$

82. The area of the region  $A = \left\{ (x, y) : |\cos x - \sin x| \leq y \leq \sin x, 0 \leq x \leq \frac{\pi}{2} \right\}$  is

क्षेत्र  $A = \left\{ (x, y) : |\cos x - \sin x| \leq y \leq \sin x, 0 \leq x \leq \frac{\pi}{2} \right\}$  का क्षेत्रफल है :

(A)  $\sqrt{5} - 2\sqrt{2} + 1$

(B)  $1 - \frac{3}{\sqrt{2}} + \frac{4}{\sqrt{5}}$

(C)  $\frac{3}{\sqrt{5}} - \frac{3}{\sqrt{2}} + 1$

(D)  $\sqrt{5} + 2\sqrt{2} - 4.5$

**Ans.** A

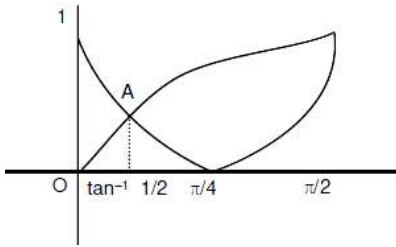
**Sol.** 
$$y = \begin{cases} \cos x - \sin x & : 0 \leq x \leq \frac{\pi}{4} \\ \sin x - \cos x & : \frac{\pi}{4} < x \leq \frac{\pi}{2} \end{cases}$$

$$y^1 = \begin{cases} -\sin x - \cos x & : 0 \leq x \leq \frac{\pi}{4} \\ \cos x + \sin x & : \frac{\pi}{4} < x \leq \frac{\pi}{2} \end{cases}$$

For pt. A

$$\cos x - \sin x = \sin x$$

$$\tan x = \frac{1}{2}$$



$$\text{Let } x = \tan^{-1} \frac{1}{2}$$

$$\begin{aligned} \text{Area} &= \int_{\tan^{-1} 1/2}^{\pi/4} (\sin x - (\cos x - \sin x)) dx + \int_{\pi/4}^{\pi/2} (\sin x - \sin x + \cos x) dx \\ &= (-2 \cos x - \sin x) \Big|_{\tan^{-1} 1/2}^{\pi/4} + (\sin x) \Big|_{\pi/4}^{\pi/2} \\ &= \left( \frac{-2}{\sqrt{2}} - \frac{1}{\sqrt{2}} \right) - (-2 \cos x - \sin x) + \left( 1 - \frac{1}{\sqrt{2}} \right) \\ &= -\sqrt{2} - \frac{1}{\sqrt{2}} + 2 \cos x + \sin x + 1 - \frac{1}{\sqrt{2}} = -2\sqrt{2} + 1 + \frac{2 \times 2}{\sqrt{5}} + \frac{1}{\sqrt{5}} \\ &= \sqrt{5} + 1 - 2\sqrt{2} \end{aligned}$$

83. Let  $y = y(t)$  be a solution of the differential equation  $\frac{dy}{dt} + \alpha y = \gamma e^{-\beta t}$

where,  $\alpha > 0, \beta > 0$  and  $\gamma > 0$ . Then  $\lim_{t \rightarrow \infty} y(t)$

(A) is  $-1$  (B) is  $0$  (C) does not exist (D) is  $1$

$$\text{माना अवकल समीकरण } \frac{dy}{dt} + \alpha y = \gamma e^{-\beta t}$$

जहाँ  $\alpha > 0, \beta > 0$  तथा  $\gamma > 0$  है, का हल  $y = y(t)$  है। तो  $\lim_{t \rightarrow \infty} y(t)$

(A)  $-1$  है (B)  $0$  है (C) का अस्तित्व नहीं है (D)  $1$  है

**Ans.** B

$$\text{Sol. I.F.} = e^{\int \alpha dt} = e^{\alpha t}$$

$$\text{Solution } y \cdot e^{\alpha t} = \int e^{\alpha t} \cdot \gamma \cdot e^{-\beta t} dt + C$$

$$\Rightarrow ye^{\alpha t} = \gamma \int e^{(\alpha-\beta)t} dt + c$$

$$\Rightarrow y(t) \cdot e^{\alpha t} = \gamma \frac{e^{(\alpha-\beta)t}}{(\alpha-\beta)} + c$$

$$\Rightarrow y(t) = \gamma \frac{e^{-\beta t}}{(\alpha-\beta)} + ce^{-\alpha t}$$

$$\lim_{t \rightarrow \infty} y(t) = \lim_{t \rightarrow \infty} \left( \frac{\gamma \cdot e^{-\beta t}}{(\alpha-\beta)} - c \cdot e^{-\alpha t} \right)$$

$$= 0 - 0 = 0$$

84. The minimum value of the function  $f(x) = \int_0^2 e^{|x-t|} dt$  is :

फलन  $f(x) = \int_0^2 e^{|x-t|} dt$  का निम्नतम मान है :

(A) 2

(B)  $e(e-1)$

(C)  $2(e-1)$

(D)  $2e-1$

**Ans.** C

**Sol.** For  $x > 2$

$$f(x) = \int_0^2 e^{x-t} dt$$

$$= e^x \left( -e^{-t} \right) \Big|_0^2$$

$$= e^x (1 - e^{-2})$$

For  $x < 0$

$$f(x) = \int_0^2 e^{t-x} dt = e^{-x} e^t \Big|_0^2 = e^{-x} (e^2 - 1)$$

For  $0 \leq x \leq 2$

$$f(x) = \int_0^x e^{x-t} dt + \int_x^2 e^{t-x} dt$$

$$= -e^x e^{-t} \Big|_0^x + e^{-x} e^t \Big|_x^2$$

$$\Rightarrow -e^x (e^{-x} - 1) + e^{-x} (e^2 - e^x)$$

$$\Rightarrow -1 + e^x + e^{2-x} - 1$$

$$= e^{2-x} + e^x - 2$$

$$f(x) = \begin{cases} e^x(1-e^{-2}); & x > 2 \\ e^{2-x} + e^x - 2; & 0 \leq x \leq 2 \\ e^{-x}(e^2 - 1); & x < 0 \end{cases}$$

For  $x > 2$

$$f(x)_{\min} = e^2 - 1$$

For  $0 \leq x \leq 2$

$$f'(x) = -e^{2-x} + e^x = 0 \Rightarrow e^x = e^{2-x} \Rightarrow e^{2x} = e^2 \Rightarrow x = 1$$

$$f(x) = 2e - 2 = 2(e - 1)$$

For  $x < 0$

$$f(x)_{\min} = e^2 - 1$$

85. Let  $P(x) = x^2 + bx + c$  be a quadratic polynomial with real coefficients such that  $\int_0^1 P(x)dx = 1$  and  $P(x)$  leaves remainder 5 when it is divided by  $(x-2)$ . That the value of  $9(b+c)$  is equal to :

माना वास्तविक गुणांकों का एक द्विघातीय बहुपद  $P(x) = x^2 + bx + c$  इस प्रकार है कि  $\int_0^1 P(x)dx = 1$  है तथा  $P(x)$  को

$(x-2)$  से विभाजित करने पर शेषफल 5 आता है। तो  $9(b+c)$  का मान बराबर है :

- (A) 7 (B) 9 (C) 11 (D) 15

Ans. A

Sol.  $P(x) = x^2 + bx + c$

$$\int_0^1 (x^2 + bx + c)dx = 1$$

$$\frac{1}{3} + \frac{b}{2} + c = 1$$

$$3b + 6c = 4 \dots (i)$$

$$P(2) = 5$$

$$4 + 2b + c = 5$$

$$2b + c = 1 \dots (ii)$$

From (i) and (ii)

$$b = \frac{2}{9}, c = \frac{5}{9}$$

$$b + c = \frac{7}{9}$$

$$9(b + c) = 7$$

86. Let  $n \geq 2$  be a natural number and  $0 < \theta < \pi/2$ . Then  $\int \frac{(\sin^n \theta - \sin \theta)^{\frac{1}{n}} \cos \theta}{\sin^{n+1} \theta} d\theta$  is equal to : (where C is a constant of integration)

माना  $n \geq 2$  एक प्राकृत संख्या है तथा  $0 < \theta < \pi/2$  है, तो  $\int \frac{(\sin^n \theta - \sin \theta)^{\frac{1}{n}} \cos \theta}{\sin^{n+1} \theta} d\theta$  बराबर है : (जहाँ C एक समाकलन अचर है)

$$(A) \frac{n}{n^2 - 1} \left( 1 - \frac{1}{\sin^{n-1} \theta} \right)^{\frac{n+1}{n}} + C$$

$$(B) \frac{n}{n^2 - 1} \left( 1 + \frac{1}{\sin^{n-1} \theta} \right)^{\frac{n+1}{n}} + C$$

$$(C) \frac{n}{n^2 + 1} \left( 1 - \frac{1}{\sin^{n-1} \theta} \right)^{\frac{n+1}{n}} + C$$

$$(D) \frac{n}{n^2 - 1} \left( 1 - \frac{1}{\sin^{n+1} \theta} \right)^{\frac{n+1}{n}} + C$$

Ans. A

Sol.  $\int \frac{(\sin^n \theta - \sin \theta)^{\frac{1}{n}} \cos \theta}{\sin^{n+1} \theta} d\theta$

$$\int \sin \theta \frac{\left( 1 - \frac{1}{\sin^{n-1} \theta} \right)^{\frac{1}{n}}}{\sin^{n+1} \theta} \cos \theta d\theta$$

$$1 - \frac{1}{\sin^{n-1} \theta} = t$$

$$(n-1) \frac{1}{\sin^n \theta} \cos \theta d\theta$$

$$\int \frac{(t)^{1/n}}{(n-1)} dt = \frac{1}{(n-1)} \cdot \frac{(t)^{\frac{1}{n}+1}}{\frac{1}{n}+1} + C$$

$$= \left( \frac{n}{n^2 - 1} \right) \left( 1 - \frac{1}{\sin^{n-1} \theta} \right)^{\frac{n+1}{n}} + C$$

87. Let  $\alpha x = \exp(x^\beta y^\gamma)$  be the solution of the differential equation  $2x^2y \, dy - (1 - xy^2)dx = 0$ ,  $x > 0$ ,  $y(2) = \sqrt{\log_e 2}$ . Then  $\alpha + \beta - \gamma$  equals :

माना अवकल समीकरण  $2x^2y \, dy - (1 - xy^2)dx = 0$ ,  $x > 0$ ,  $y(2) = \sqrt{\log_e 2}$  का हल  $\alpha x = \exp(x^\beta y^\gamma)$  है। तो  $\alpha + \beta - \gamma$  :

- (A) -1 (B) 1 (C) 0 (D) 3

**Ans.** B

**Sol.**  $\alpha x = e^{x^\beta \cdot y^\gamma}$

$$2x^2y \frac{dy}{dx} = 1 - xy^2 \quad y^2 = t$$

$$x^2 \frac{dt}{dx} = 1 - xt$$

$$\frac{dt}{dx} + \frac{t}{x} = \frac{1}{x^2}$$

$$\text{I.F.} = e^{\ln x} = x$$

$$t(x) = \int \frac{1}{x^2} \cdot x dx$$

$$y^2 \cdot x = \ln x + C$$

$$\therefore 2 \cdot \ln 2 = \ln 2 + C$$

$$\therefore C = \ln 2$$

$$\text{Hence, } xy^2 = \ln 2x$$

$$\therefore 2x = e^{x \cdot y^2}$$

$$\text{Hence } \alpha = 2, \beta = 1, \gamma = 2$$

88. The solution curve of the differential equation,  $(1 + e^{-x})(1 + y^2) \frac{dy}{dx} = y^2$ , which passes through the point (0, 1), is :

अवकल समीकरण  $(1 + e^{-x})(1 + y^2) \frac{dy}{dx} = y^2$  का हल वक्र, जो बिन्दु (0, 1) से होकर जाता है, है :

(A)  $y^2 + 1 = y \left( \log_e \left( \frac{1 + e^{-x}}{2} \right) + 2 \right)$  (B)  $y^2 + 1 = y \left( \log_e \left( \frac{1 + e^x}{2} \right) + 2 \right)$

(C)  $y^2 = 1 + y \log_e \left( \frac{1 + e^{-x}}{2} \right)$  (D)  $y^2 = 1 + y \log_e \left( \frac{1 + e^x}{2} \right)$

Ans. D

Sol.  $\frac{1+y^2}{y^2} dy = \frac{dx}{1+e^{-x}}$

$$\Rightarrow \int \left( \frac{1}{y^2} + 1 \right) dy = \int \frac{e^x}{e^x + 1} dx$$

$$\Rightarrow \frac{-1}{y} + y = \ln(e^x + 1) + C$$

$$y(0) = 1$$

$$\Rightarrow -1 + 1 = \ln 2 + C$$

$$C = -\ln 2$$

$$y - \frac{1}{y} = \ln(e^x + 1) - \ln 2$$

$$\Rightarrow y^2 = 1 + y \ln \left( \frac{e^x + 1}{2} \right)$$

89. Given :  $f(x) = \begin{cases} x & , 0 \leq x < \frac{1}{2} \\ \frac{1}{2} & , x = \frac{1}{2} \\ 1-x & , \frac{1}{2} < x \leq 1 \end{cases}$  and  $g(x) = \left(x - \frac{1}{2}\right)^2, x \in \mathbf{R}$ . Then the area (in sq. units) of the

region bounded by the curves,  $y = f(x)$  and  $y = g(x)$  between the lines,  $2x = 1$  and  $2x = \sqrt{3}$ , is :

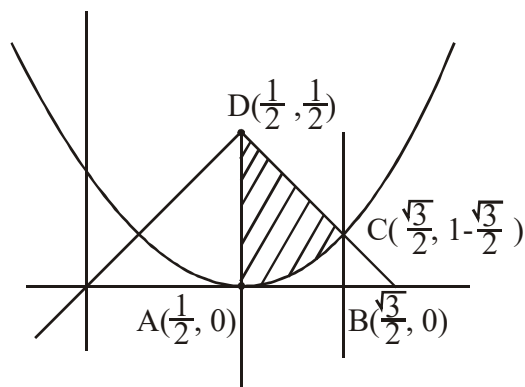
दिया है  $f(x) = \begin{cases} x & , 0 \leq x < \frac{1}{2} \\ \frac{1}{2} & , x = \frac{1}{2} \\ 1-x & , \frac{1}{2} < x \leq 1 \end{cases}$  तथा  $g(x) = \left(x - \frac{1}{2}\right)^2, x \in \mathbf{R}$ . तो रेखाओं  $2x = 1$  तथा  $2x = \sqrt{3}$  के बीच ,

वक्रों  $y = f(x)$  तथा  $y = g(x)$  द्वारा प्रतिबद्ध क्षेत्र का क्षेत्रफल (वर्ग इकाइयों में) है :

(A)  $\frac{\sqrt{3}}{4} - \frac{1}{3}$  (B)  $\frac{1}{3} + \frac{\sqrt{3}}{4}$  (C)  $\frac{1}{2} + \frac{\sqrt{3}}{4}$  (D)  $\frac{1}{2} - \frac{\sqrt{3}}{4}$

Ans. A

Sol.



$$\text{Required area} = \text{Area of trapezium ABCD} - \int_{\frac{1}{2}}^{\frac{\sqrt{3}}{2}} \left( \frac{x-1}{2} \right)^2 dx$$

$$\text{Area of trapezium } m = \frac{1}{2} (\text{Sum of parallel sides}) (h)$$

$$= \frac{1}{2} \left( \frac{1}{2} + 1 - \frac{\sqrt{3}}{2} \right) \left( \frac{\sqrt{3}}{2} - \frac{1}{2} \right)$$

$$= \frac{1}{2} \left( \frac{3}{2} - \frac{\sqrt{3}}{2} \right) \left( \frac{\sqrt{3}}{2} - \frac{1}{2} \right)$$

$$= \frac{1}{2} \left[ \frac{3\sqrt{3}}{4} - \frac{3}{4} - \frac{3}{4} + \frac{\sqrt{3}}{4} \right] = \frac{1}{2} \left( \sqrt{3} - \frac{3}{2} \right)$$

$$= \frac{\sqrt{3}}{2} - \frac{3}{4}$$

$$\int_{\frac{1}{2}}^{\frac{\sqrt{3}}{2}} \left( x - \frac{1}{2} \right)^2 dx = \frac{1}{3} \left( x - \frac{1}{2} \right)^3 \Big|_{\frac{1}{2}}^{\frac{\sqrt{3}}{2}}$$

$$= \frac{1}{3} \left( \left( \frac{\sqrt{3}}{2} - \frac{1}{2} \right)^3 \right) = \frac{1}{3} \times \frac{1}{8} (\sqrt{3} - 1)^3$$

$$= \frac{1}{24} (3\sqrt{3} - 1 + 3 \times \sqrt{3} - 9)$$

$$= \frac{1}{24}(6\sqrt{3}-10) = \frac{1}{12}(3\sqrt{3}-5)$$

Final Ans. 
$$= \frac{\sqrt{3}}{2} - \frac{3}{4} - \frac{1}{12}(3\sqrt{3}-5)$$

$$= \frac{\sqrt{3}}{2} - \frac{3}{4} - \frac{\sqrt{3}}{4} + \frac{5}{12}$$

$$= \frac{\sqrt{3}}{4} + \frac{5}{12} - \frac{3}{4} = \frac{\sqrt{3}}{4} + \frac{5-9}{12}$$

$$= \frac{\sqrt{3}}{4} - \frac{4}{12} = \frac{\sqrt{3}}{4} - \frac{1}{3}$$

90. The integral  $\int \frac{2x^3-1}{x^4+x} dx$  is equal to :

(Here C is a constant of integration)

समाकल  $\int \frac{2x^3-1}{x^4+x} dx$  बराबर है :

(यहाँ C समाकलन अचर है)

(A)  $\log_e \left| \frac{x^3+x}{x} \right| + C$

(B)  $\frac{1}{2} \log_e \frac{(x^3+1)^2}{|x^3|} + C$

(C)  $\frac{1}{2} \log_e \frac{|x^3+1|}{x^2} + C$

(D)  $\log_e \frac{|x^3+1|}{x^2} + C$

Ans. A

sol. 
$$I = \int \frac{(2x^3-1)dx}{x^4+x} = \int \frac{(2x-x^{-2})dx}{x^2+x^{-1}}$$

Put  $x^2 + x^{-1} = t$

$$(2x - x^{-2})dx = dt$$

$$I = \int \frac{dt}{t} = \ln |t| + c$$

$$= \ln |x^2 + x^{-1}| + c$$

$$= \ln \left| \frac{x^3+1}{x} \right| + c$$

91. Consider a region  $R = \{(x, y) \in \mathbb{R}^2 : x^2 \leq y \leq 2x\}$ . If a line  $y = \alpha$  divides the area of region R into two equal

parts, then which of the following is true ?

एक क्षेत्र  $R = \{(x, y) \in \mathbb{R}^2 : x^2 \leq y \leq 2x\}$  पर विचार कीजिए। यदि एक सरल रेखा  $y = \alpha$ , क्षेत्र R के क्षेत्रफल को दो बराबर भागों में बांटती है, तो निम्न में से कौन - सा सत्य है?

(A)  $3\alpha^2 - 8\alpha + 8 = 0$

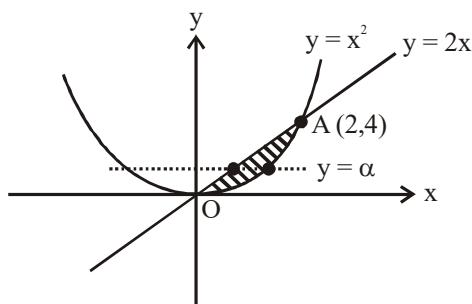
(B)  $\alpha^3 - 6\alpha^2 + 16 = 0$

(C)  $3\alpha^2 - 8\alpha^{\frac{3}{2}} + 8 = 0$

(D)  $\alpha^3 - 6\alpha^{\frac{3}{2}} - 16 = 0$

**Ans.** C

**Sol.**  $\because x^2 \leq y \leq 2x$



For point of intersection O & A

$$x^2 = 2x \quad \therefore x = 0, x = 2$$

$$O = (0, 0) \quad A = (2, 4)$$

We have

$$\int_0^\alpha \left( \sqrt{y} - \frac{y}{2} \right) dy = \int_\alpha^4 \left( \sqrt{y} - \frac{y}{2} \right) dy$$

$$\Rightarrow \left( \frac{2}{3} y^{3/2} - \frac{y^2}{4} \right)_0^\alpha = \left( \frac{2}{3} y^{3/2} - \frac{y^2}{4} \right)_\alpha^4$$

$$\Rightarrow \frac{2}{3} \alpha^{3/2} - \frac{1}{4} \alpha^2 = \left( \frac{2}{3} \cdot 8 - 4 \right) - \left( \frac{2}{3} \alpha^{3/2} - \frac{\alpha^2}{4} \right)$$

$$\Rightarrow \frac{4}{3} \alpha^{3/2} - \frac{1}{2} \alpha^2 = \frac{4}{3}$$

$$8\alpha^{3/2} - 3\alpha^2 = 8$$

92. If the curve  $y = y(x)$  is the solution of the differential equation  $2 \left( x^2 + x^{\frac{5}{4}} \right) dy - y \left( x + x^{\frac{1}{4}} \right) dx = 2x^{\frac{9}{4}} dx, x >$

0 which passes through the point  $\left(1, 1 - \frac{4}{3} \log_e 2\right)$ , then the value of  $y(16)$  is equal to:

यदि अवकल समीकरण  $2\left(x^2 + x^{\frac{5}{4}}\right)dy - y\left(x + x^{\frac{1}{4}}\right)dx = 2x^{\frac{9}{4}}dx$ ,  $x > 0$  का हल वक्र  $y = (x)$  हैं, जो बिन्दु

$\left(1, 1 - \frac{4}{3} \log_e 2\right)$  से होकर जाता है तो  $y(16)$  का मान बराबर है :

- (A)  $\left(\frac{31}{3} + \frac{8}{3} \log_e 3\right)$  (B)  $4\left(\frac{31}{3} + \frac{8}{3} \log_e 3\right)$  (C)  $4\left(\frac{31}{3} - \frac{8}{3} \log_e 3\right)$  (D)  $\left(\frac{31}{3} - \frac{8}{3} \log_e 3\right)$

Ans. C

Sol.  $2\left(x^2 + x^{\frac{5}{4}}\right)dy - y\left(x + x^{\frac{1}{4}}\right)dx = 2x^{\frac{9}{4}}$

$$\frac{dy}{dx} - \frac{y}{2} \left( \frac{x + x^{1/4}}{x^2 + x^{5/4}} \right) = \frac{x^{9/4}}{x^2 + x^{5/4}}$$

$$\frac{dy}{dx} - \frac{y}{2} \left( \frac{1}{x} \right) = \frac{x^{9/4}}{x^2 + x^{5/4}}$$

$$I. F = e^{\int -\frac{1}{2x} dx}$$

$$= e^{-\frac{1}{2} \ln x} = \frac{1}{\sqrt{x}}$$

$$y \left( \frac{1}{\sqrt{x}} \right) = \int \frac{1}{\sqrt{x}} \times \frac{x^{9/4}}{x^2 + x^{5/4}} dx$$

$$\frac{y}{\sqrt{x}} = \int \frac{x^{7/4} dx}{x(x + x^{1/4})}$$

$$\frac{y}{\sqrt{x}} = \int \frac{x^{3/4}}{x + x^{1/4}} dx$$

$$x = y^4$$

$$dx = 4y^3 dy$$

$$\frac{y}{\sqrt{x}} = \int \frac{4y^3 \times y^3}{y^4 + y} dy$$

$$\frac{y}{\sqrt{x}} = 4 \int \frac{y^5 dy}{y^3 + 1}$$

$$y^3 = t$$

$$3y^2 dy = dt$$

$$y^2 dy = \frac{1}{3} dt$$

$$\frac{y}{\sqrt{x}} = \frac{4}{3} \int \frac{t}{1+t} dt$$

$$\frac{y}{\sqrt{x}} = \frac{4}{3} (t - \ln(1+t)) + C$$

$$\frac{y}{\sqrt{x}} = \frac{4}{3} (y^3 - \ln(1+y^3)) + C$$

$$\frac{y}{\sqrt{x}} = \frac{4}{3} \left( x^{\frac{3}{4}} - \ln\left(1+x^{\frac{3}{4}}\right) \right) + C$$

$$\text{It passes through } \left(1, 1 - \frac{4}{3} \ln 2\right)$$

$$1 - \frac{4}{3} \ln 2 = \frac{4}{3} (1 - \ln 2) + C$$

$$C = -\frac{1}{3}$$

$$\frac{y}{\sqrt{x}} = \frac{4}{3} \left( x^{3/4} - \ln(1+x^{3/4}) \right) - \frac{1}{3}$$

$$x = 16$$

$$\frac{y}{4} = \frac{4}{3} (8 - \ln(9)) - \frac{1}{3}$$

$$\frac{y}{4} = \frac{32}{3} - \frac{8}{3} \ln 3 - \frac{1}{3}$$

$$\frac{y}{4} = \frac{31}{3} - \frac{8}{3} \ln 3$$

$$y = 4 \left( \frac{31}{3} - \frac{8}{3} \ln 3 \right)$$

**INTEGER TYPE QUESTIONS**

93. If  $\int_0^{\pi} \frac{5^{\cos x} (1 + \cos x \cos 3x + \cos^2 x + \cos^3 x \cos 3x) dx}{1 + 5^{\cos x}} = \frac{k\pi}{16}$ , then k is equal to \_\_\_\_\_.

यदि  $\int_0^{\pi} \frac{5^{\cos x} (1 + \cos x \cos 3x + \cos^2 x + \cos^3 x \cos 3x) dx}{1 + 5^{\cos x}} = \frac{k\pi}{16}$  है, तो k बराबर है।

**Ans.** 26

**Sol.** 
$$I = \int_0^{\pi} \frac{5^{\cos x} (1 + \cos x \cos 3x + \cos^2 x + \cos^3 x \cos 3x) dx}{1 + 5^{\cos x}}$$

$$= 2 \int_0^{\pi/2} (1 + \cos x \cos 3x + \cos^2 x + \cos^3 x \cos 3x) dx$$

$$= \int_0^{\pi/2} 2 + \cos 4x + \cos 2x + 1 + \cos 2x + 2 \cos 3x \left( \frac{\cos 3x + 3 \cos x}{4} \right) dx$$

$$= \int_0^{\pi/2} 3 dx + 0 + 0 + 0 + \int_0^{\pi/2} \frac{1}{2} (\cos^2 3x + 3 \cos x \cos 3x) dx$$

$$= \frac{3\pi}{2} + \frac{1}{4} \int_0^{\pi/2} (1 + \cos 6x + 3(\cos 4x + \cos 2x)) dx$$

$$= \frac{3\pi}{4} + \frac{1}{4} \cdot \frac{\pi}{2} = \frac{13\pi}{8} = \frac{k\pi}{16}$$

$$\Rightarrow k = 26$$

94. Let  $I_n = \int_1^e x^{19} (\log |x|)^n dx$ , where  $n \in \mathbb{N}$ . If  $(20)I_{10} = \alpha I_9 + \beta I_8$ , for natural numbers  $\alpha$  and  $\beta$ , then

$\alpha - \beta$  equals to \_\_\_\_.

माना  $I_n = \int_1^e x^{19} (\log |x|)^n dx$  है जहाँ  $n \in \mathbb{N}$  है। यदि धन पूर्णाकों  $\alpha$  तथा  $\beta$  के लिए  $(20)I_{10} = \alpha I_9 + \beta I_8$  है, तो

$\alpha - \beta$  बराबर है \_\_\_\_।

Ans. 1

Sol.  $I_n = \int_1^e x^{19} (\log x)^n dx$

$$I_n = \int_1^e (\log x)^n x^{19} dx$$

$$I_n = \left[ \frac{(\log x)^n x^{20}}{20} \right]_1^e - \int_1^e \frac{n(\log x)^{n-1}}{x} \cdot \frac{x^{20}}{20} dx$$

$$I_n = \frac{e^{20}}{20} - \frac{n}{20} \int_1^e (\log x)^{n-1} x^{19} dx$$

$$20 I_n = e^{20} - n I_{n-1}$$

$$20 I_n + n I_{n-1} = e^{20}$$

$$n = 10$$

$$20 I_{10} + 10 I_9 = e^{20} \dots\dots\dots(1)$$

$$n = 9$$

$$20 I_9 + 9 I_8 = e^{20} \dots\dots\dots(2)$$

By (1) & (2)

$$20 I_{10} + 10 I_9 = 20 I_9 + 9 I_8$$

$$20 I_{10} = 10 I_9 + 9 I_8$$

By Comparision

$$\alpha = 10$$

$$\beta = 9$$

$$\alpha - \beta = 1$$

95. Let A be the area bounded by the curve  $y = x|x - 3|$ , the x-axis and the ordinates  $x = -1$  and  $x = 2$ . Then 12A is equal to \_\_\_\_\_.

माना वक्र  $y = x|x - 3|$ , x-अक्ष तथा कोटियों  $x = -1$  व  $x = 2$  से घिरे क्षेत्र का क्षेत्रफल A है। तब 12A का मान है \_\_\_\_\_

Ans. 62

Sol.  $A = \int_{-1}^0 (x^2 - 3x) dx + \int_0^2 (3x - x^2) dx$

$$\Rightarrow A = \left. \frac{x^3}{3} - \frac{3x^2}{2} \right|_{-1}^0 + \left. \frac{3x^2}{2} - \frac{x^3}{3} \right|_0^2$$

$$\Rightarrow A = \frac{11}{6} + \frac{10}{3} = \frac{31}{6}$$

$$\therefore 12A = 62$$

96. If  $y = y(x)$  is the solution of the equation  $e^{\sin y} \cos y \frac{dy}{dx} + e^{\sin y} \cos x = \cos x, y(0) = 0$ ; then

$$1 + y\left(\frac{\pi}{6}\right) + \frac{\sqrt{3}}{2}y\left(\frac{\pi}{3}\right) + \frac{1}{\sqrt{2}}y\left(\frac{\pi}{4}\right) \text{ is equal to :}$$

यदि समीकरण  $e^{\sin y} \cos y \frac{dy}{dx} + e^{\sin y} \cos x = \cos x, y(0) = 0$  का हल  $y = y(x)$  है, तो

$$1 + y\left(\frac{\pi}{6}\right) + \frac{\sqrt{3}}{2}y\left(\frac{\pi}{3}\right) + \frac{1}{\sqrt{2}}y\left(\frac{\pi}{4}\right) \text{ बराबर है } \underline{\hspace{2cm}} \text{ ।}$$

Ans. 1

Sol.  $e^{\sin y} \cos y \frac{dy}{dx} + e^{\sin y} \cos x = \cos x$

Let  $e^{\sin y} = t$

$$\Rightarrow e^{\sin y} \cos y \frac{dy}{dx} = \frac{dt}{dx}$$

$$\Rightarrow \frac{dt}{dx} + t \cos x = \cos x$$

Integrating Factors

$$\text{I.F.} = e^{\int \cos x dx} = e^{\sin x}$$

$$\Rightarrow \int d(t \cdot e^{\sin x}) dx = \int e^{\sin x} \cos x dx$$

$$\Rightarrow t \cdot e^{\sin x} = e^{\sin x} + c$$

$$\Rightarrow e^{\sin y} \cdot e^{\sin x} = e^{\sin x} + c$$

$$x = 0; y = 0 \Rightarrow c = 0$$

$$\Rightarrow e^{\sin y} \cdot e^{\sin x} = e^{\sin x}$$

$$\Rightarrow e^{\sin y} = 1 \Rightarrow \sin y = 0 \Rightarrow y = 0$$

$$\Rightarrow 1 + y\left(\frac{\pi}{6}\right) + \frac{\sqrt{3}}{2}y\left(\frac{\pi}{3}\right) + \frac{1}{\sqrt{2}}y\left(\frac{\pi}{4}\right) = 1$$

97. If  $\int \sqrt{\sec 2x - 1} dx = \alpha \log_e \left| \cos 2x + \beta + \sqrt{\cos 2x \left( 1 + \cos \frac{1}{\beta} x \right)} \right| + \text{constant}$ , then  $\beta - \alpha$  is equal to \_\_\_\_\_.

यदि  $\int \sqrt{\sec 2x - 1} dx = \alpha \log_e \left| \cos 2x + \beta + \sqrt{\cos 2x \left( 1 + \cos \frac{1}{\beta} x \right)} \right| + \text{constant}$ , तो  $\beta - \alpha$  बराबर है।

**Ans.** 1

**Sol.** 
$$\int \sqrt{\sec 2x - 1} dx = \int \sqrt{\frac{1 - \cos 2x}{\cos 2x}} dx = \sqrt{2} \int \frac{\sin x}{\sqrt{2 \cos^2 x - 1}} dx$$

put  $\cos x = t \Rightarrow -\sin x dx = dt$

$$= -\sqrt{2} \int \frac{dt}{\sqrt{2t^2 - 1}} = -\ln \left| \sqrt{2} \cos x + \sqrt{\cos 2x} \right| + c = -\frac{1}{2} \ln \left| \cos 2x + \frac{1}{2} + \sqrt{\cos 2x} \cdot \sqrt{1 + \cos 2x} \right| + c$$

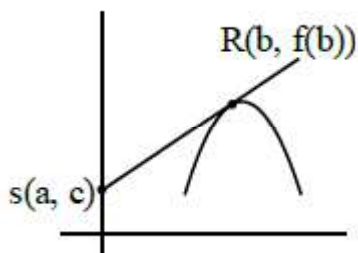
$$\therefore \beta = \frac{1}{2}, \alpha = -\frac{1}{2} \Rightarrow \beta - \alpha = 1$$

98. Let a curve  $y = f(x)$ ,  $x \in (0, \infty)$  pass through the points  $P\left(1, \frac{3}{2}\right)$  and  $Q\left(a, \frac{1}{2}\right)$ . If the tangent at any point  $R(b, f(b))$  to the given curve cuts the y-axis at the point  $S(0, c)$  such that  $bc = 3$ , then  $(PQ)^2$  is equal to \_\_\_\_\_.

माना एक वक्र  $y = f(x)$ ,  $x \in (0, \infty)$  बिंदुओं  $P\left(1, \frac{3}{2}\right)$  तथा  $Q\left(a, \frac{1}{2}\right)$  से होकर जाता है। यदि दिए गए वक्र के किसी भी बिंदु  $R(b, f(b))$  पर स्पर्श रेखा y-अक्ष को बिंदु  $S(0, c)$  पर काटती है जबकि  $bc = 3$  तो  $(PQ)^2$  बराबर है \_\_\_\_\_

**Ans.** 5

**Sol.**



Equation of tangent at  $R(b, f(2))$  is

$$y - f(b) = f'(b) \cdot (x - b)$$

which passes through  $(0, c)$

$$\Rightarrow c - f(b) = f'(b) \cdot (-b)$$

$$\Rightarrow \frac{3}{b} - f(b) = f'(b) \cdot (-b)$$

$$\Rightarrow bf'(b) - f(b) = -\frac{3}{b}$$

$$\Rightarrow \frac{bf'(b) - f(b)}{b^2} = -\frac{3}{b^3}$$

$$\Rightarrow d\left(\frac{f(b)}{b}\right) = -\frac{3}{b^3} \Rightarrow \frac{f(b)}{b} = \frac{3}{2b^2} + \lambda$$

Which passes through  $(1, 3/2)$

$$\Rightarrow \frac{3}{2} = \frac{3}{2} + \lambda \Rightarrow \lambda = 0$$

$$\Rightarrow f(b) = \frac{3}{2b}$$

$$f(a) = \frac{1}{2} \Rightarrow \frac{1}{2} = \frac{3}{2b} \Rightarrow b = 3$$

$$\Rightarrow c = 1 \Rightarrow Q(3, 1/2)$$

$$\Rightarrow PQ^2 = 2^2 + (1)^2 = 5$$

99. Let  $A$  be the area of the region  $\{(x, y) : y \geq x^2, y \geq (1-x)^2, y \leq 2x(1-x)\}$ . Then  $540A$  is equal to \_\_\_\_\_.

माना क्षेत्र  $\{(x, y) : y \geq x^2, y \geq (1-x)^2, y \leq 2x(1-x)\}$  का क्षेत्रफल  $A$  है। तब  $540A$  का मान बराबर है \_\_\_\_\_

**Ans.** 25

**Sol.**

$$A = 2 \int_{\frac{1}{3}}^{\frac{1}{2}} (2x - 2x^2 - (1-x)^2) dx$$

$$= 2 \left[ 2x^2 - x^3 - x \right]_{1/3}^{1/2}$$

$$\text{So } A = \frac{5}{108} \Rightarrow 540A = \frac{5}{108} \times 540 = 25$$

100. If  $I_{m,n} = \int_0^1 x^{m-1} (1-x)^{n-1} dx$ , for  $m, n \geq 1$ , and  $\int_0^1 \frac{x^{m-1} + x^{n-1}}{(1+x)^{m+n}} dx = \alpha I_{m,n}, \alpha \in \mathbb{R}$ , then  $\alpha$  equals \_\_\_\_\_.

यदि  $I_{m,n} = \int_0^1 x^{m-1} (1-x)^{n-1} dx$ ,  $m, n \geq 1$ , तथा  $\int_0^1 \frac{x^{m-1} + x^{n-1}}{(1+x)^{m+n}} dx = \alpha I_{m,n}, \alpha \in \mathbb{R}$  हैं, तो  $\alpha$  बराबर है \_\_\_\_\_।

Ans. 1

Sol.  $I_{m,n} = \int_0^1 x^{m-1} (1-x)^{n-1} dx$

$$\text{put } x = \frac{1}{1+t} \quad \therefore t = \frac{1}{x} - 1$$

$$dx = \frac{-1}{(1+t)^2} dt$$

$$= \int_{\infty}^0 \frac{1}{(1+t)^{m-1}} \left(1 - \frac{1}{1+t}\right)^{n-1} \left(\frac{-1}{(1+t)^2}\right) dt$$

$$= - \int_{\infty}^0 \frac{t^{n-1}}{(1+t)^{m-1} (1+t)^{n-1}} \left(\frac{1}{(1+t)^2}\right) dt$$

$$= \int_0^{\infty} \frac{t^{n-1}}{(1+t)^{m-1} (1+t)^{n-1} (1+t)^2} dt$$

$$= \int_0^{\infty} \frac{t^{n-1}}{(1+t)^{m+n}} dt$$

$$= \int_0^1 \frac{t^{n-1}}{(1+t)^{m+n}} dt + \int_1^{\infty} \frac{t^{n-1}}{(1+t)^{m+n}} dt$$

$$I_1 = \int_1^{\infty} \frac{t^{n-1}}{(1+t)^{m+n}} dt$$

$$\text{put } t = \frac{1}{z} \quad \therefore dt = \frac{-1}{z^2} dz$$

$$= \int_1^0 \frac{1}{z^{n-1} \left(1 + \frac{1}{z}\right)^{m+n}} \cdot \left(\frac{-1}{z^2}\right) dz$$

$$= - \int_1^0 \frac{1}{z^{n-1} \frac{(z+1)^{m+n}}{z^{m+n}} z^2} dz$$

$$I_1 = \int_0^1 \frac{z^{m-1}}{(1+z)^{m+n}} dz$$

$$I_1 = \int_0^1 \frac{t^{m-1}}{(1+t)^{m+n}} dt$$

$$I_{m,n} = \int_0^1 \frac{t^{n-1}}{(1+t)^{m+n}} dt + \int_0^1 \frac{t^{m-1}}{(1+t)^{m+n}} dt$$

$$I_{m,n} = \int_0^1 \frac{t^{n-1} + t^{m-1}}{(1+t)^{m+n}} dt$$

$$I_{m,n} = \int_0^1 \frac{x^{n-1} + x^{m-1}}{(1+x)^{m+n}} dx$$

$$\therefore \alpha = 1$$